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ENUMERATIVE COMBINATORICS / LATTICE WALKS / SPECIAL FUNCTIONS

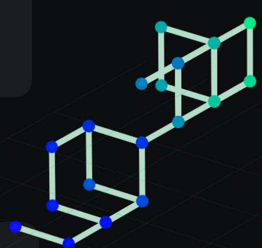
BESSEL FACTORIZATION

PROVES THE TYPE-ace WALK RECURRENCE

A candidate resolution of Mathar's conjectured D-finite recurrence for OEIS A302186

EXACT EXPONENTIAL GENERATING FUNCTION

$$F(x) = e^{2x}\frac{I_1(2x)}{x}\big(I_0(2x) + I_1(2x)\big)$$



PROOF CERTIFICATE

$$(8x+3)^3 L_6 = Q \circ L_3, \qquad L_3 F = 0$$

Metriq PRISM Laboratory
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Research Preprint MF-MATH-2026-09
Version 1.0

16 July 2026
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A Bessel-Factorization Proof of Mathar's Recurrence for Type-ace Lattice Walks

Resolution of the conjectured D-finite recurrence for OEIS A302186

Metriq PRISM Laboratory

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Research Preprint MF-MATH-2026-09 • Version 1.0 • 16 July 2026

Research status

OEIS A302186 records the number of three-dimensional walks of type *ace*. Its entry lists a five-term recurrence contributed by R. J. Mathar on 29 October 2025 and labels it conjectural. The derivation below proves that recurrence from an exact Bessel-function exponential generating function. A targeted literature search completed on 16 July 2026 did not identify a later published proof; that search is not exhaustive and does not establish priority. The argument has not undergone independent specialist review or peer review.

Abstract

Let a_n be the number of length-n nearest-neighbor walks in $\mathbb{Z}^3$ whose first coordinate stays nonnegative and returns to zero, whose second coordinate stays nonnegative with arbitrary endpoint, and whose third coordinate is unrestricted. These are Dershowitz's three-dimensional walks of type *ace*, recorded as OEIS A302186. We derive the exact exponential generating function

$$F(x) = \sum_{n\geq 0} a_n \frac{x^n}{n!} = e^{2x}\frac{I_1(2x)}{x}\big(I_0(2x) + I_1(2x)\big),$$

where I_0 and I_1 are modified Bessel functions. The first-order Bessel system yields a third-order differential equation $L_3F = 0$. We then give an exact Ore-operator factorization

$$(8x+3)^3 L_6 = Q \circ L_3,$$

where coefficient extraction from $L_6F = 0$ is precisely Mathar's conjectured recurrence. Consequently, for $n \geq 4$,

$$\begin{aligned} 0 = (n+3)(n+2)a_n &+ 4(-3n^2 - 8n - 3)a_{n-1} \\ &+ 8(4n^2 + 3n - 3)a_{n-2} + 48(n-2)^2 a_{n-3} \\ &- 144(n-2)(n-3)a_{n-4}. \end{aligned}$$

The proof is symbolic and exact. The accompanying verification independently reconstructs the coefficients from coordinate counts, checks both differential recurrences through $n = 80$, and verifies the noncommutative operator factorization exactly.

CANDIDATE MAIN RESULT

For the type-ace walk numbers a_n of OEIS A302186, Mathar's conjectured recurrence is valid for every $n \geq 4$. It follows from the Bessel EGF and the exact factorization $(8x+3)^3 L_6 = Q \circ L_3$.

MSC 2020. Primary 05A15; Secondary 05A19, 33C10, 34A25, 60G50. **Keywords.** Lattice walk; type ace; modified Bessel function; exponential generating function; D-finite sequence; Ore operator; OEIS A302186.

1 The conjectured recurrence

A walk in $\mathbb{Z}^3$ takes one of the six steps $\pm e_1, \pm e_2, \pm e_3$ at each time. Following Dershowitz's terminology, a coordinate is of type a when its one-dimensional projection stays nonnegative and returns to zero, of type c when it stays nonnegative with unrestricted endpoint, and of type e when it is unrestricted. A three-dimensional type-ace walk has one coordinate of each type.

Let a_n denote the number of such walks of length n. The first values are

$$1,\ 3,\ 11,\ 44,\ 188,\ 842,\ 3911,\ 18692,\ 91412,\ 455540, \dots.$$

The OEIS entry A302186 records the following recurrence as a conjecture.

Theorem 1.1 (Main theorem). *For every integer $n \geq 4$,*

$$\begin{aligned} 0 = {} & (n+3)(n+2)a_n + 4(-3n^2 - 8n - 3)a_{n-1} \\ & + 8(4n^2 + 3n - 3)a_{n-2} + 48(n-2)^2 a_{n-3} \\ & - 144(n-2)(n-3)a_{n-4}. \end{aligned} \tag{1}$$

The proof proceeds in three auditable stages: factor the labelled coordinate construction, eliminate the Bessel system to obtain L_3, and factor the recurrence operator L_6 through L_3.

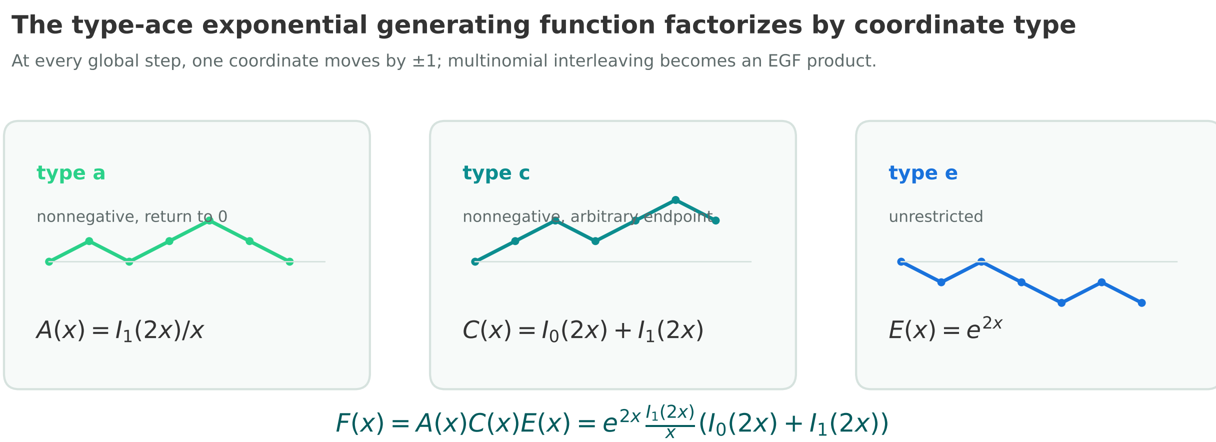

Figure 1: The coordinate constraints separate under exponential generating functions. The factors count type-a, type-c, and type-e coordinate words; their labelled interleaving reconstructs the three-dimensional walk.

2 Exact exponential generating function

Lemma 2.1 (Coordinate enumerators). *Let A_r, C_s, E_t count one-dimensional coordinate words of lengths r, s, t of types a, c, and e, respectively. Then*

$$A_{2j} = \frac{1}{j+1}\binom{2j}{j}, \qquad A_{2j+1} = 0, \qquad C_s = \binom{s}{\lfloor s/2 \rfloor}, \qquad E_t = 2^t.$$

Their exponential generating functions are

$$A(x) = \frac{I_1(2x)}{x}, \qquad C(x) = I_0(2x) + I_1(2x), \qquad E(x) = e^{2x}.$$

Proof. The type-a words are Dyck paths, hence are counted by Catalan numbers in even length and vanish in odd length. Therefore

$$A(x)=\sum_{j\ge 0}\frac{1}{j+1}\binom{2j}{j}\frac{x^{2j}}{(2j)!}=\sum_{j\ge 0}\frac{x^{2j}}{j!(j+1)!}=\frac{I_1(2x)}{x}.$$

The ballot/reflection enumeration for nonnegative prefixes gives $C_s=\binom{s}{\lfloor s/2\rfloor}$. Splitting by parity,

$$C(x)=\sum_{j\ge 0}\frac{x^{2j}}{(j!)^2}+\sum_{j\ge 0}\frac{x^{2j+1}}{j!(j+1)!}=I_0(2x)+I_1(2x).$$

An unrestricted coordinate word has two choices at every coordinate move, so $E_t=2^t$ and $E(x)=e^{2x}$. ☐

Proposition 2.2 (Bessel EGF)**.** *The type-ace walk numbers satisfy*

$$F(x):=\sum_{n\ge 0}a_n\frac{x^n}{n!}=e^{2x}\frac{I_1(2x)}{x}\big(I_0(2x)+I_1(2x)\big). \tag{2}$$

Proof. Fix coordinate-word lengths r,s,t with $r+s+t=n$. The three coordinate words can be interleaved in $\binom{n}{r,s,t}$ ways, and the coordinate constraints depend only on the corresponding projected word. Hence

$$a_n=\sum_{r+s+t=n}\binom{n}{r,s,t}A_rC_sE_t.$$

The labelled product rule for exponential generating functions and lemma 2.1 give (2). ☐

3 A third-order differential equation

Set

$$u=I_0(2x),\qquad v=I_1(2x),\qquad z=e^{-2x}F=\frac{uv+v^2}{x}.$$

The standard modified-Bessel derivative identities reduce here to

$$u'=2v,\qquad v'=2u-\frac{v}{x}. \tag{3}$$

Lemma 3.1 (Scalar elimination)**.** *The function z satisfies*

$$\begin{aligned}0&=x^2(8x+3)z'''+8x(7x+3)z''\\&\quad-4(32x^3+20x^2-12x-9)z'-4(8x+3)^2z.\end{aligned} \tag{4}$$

Proof. Introduce $w=(u^2,uv,v^2)^{\mathsf T}$. Equation (3) gives the rational first-order system

$$w'=Mw,\qquad M=\begin{pmatrix}0&4&0\\2&-x^{-1}&2\\0&4&-2x^{-1}\end{pmatrix},\qquad z=\begin{pmatrix}0&x^{-1}&x^{-1}\end{pmatrix}w. \tag{5}$$

Let $r_0=(0,x^{-1},x^{-1})$ and recursively set $r_{j+1}=r_j'+r_jM$. Then $z^{(j)}=r_jw$. The observability matrix formed from the first three rows has determinant

$$\det\begin{pmatrix}r_0\\r_1\\r_2\end{pmatrix}=-\frac{2(8x+3)}{x^5},$$

so w is eliminated uniquely over $\mathbb{Q}(x)$. Substituting that solution into $z'''=r_3w$ and clearing the denominator $x^2(8x+3)$ yields (4). This calculation uses only differentiation and rational elimination; the displayed system is a direct certificate for checking it. ☐

Conjugating (4) by $F = e^{2x}z$ replaces every occurrence of $\mathrm{D} = d/dx$ by $\mathrm{D} - 2$ and gives the next operator.

Proposition 3.2 (Annihilating operator). *Let*

$$\begin{aligned} L_3 = {} & x^2(8x+3)\mathrm{D}^3 - 2x(24x^2 - 19x - 12)\mathrm{D}^2 \\ & - 4(8x^3 + 67x^2 + 12x - 9)\mathrm{D} \\ & + 4(48x^3 + 26x^2 - 48x - 27). \end{aligned}$$

Then $L_3 F = 0$.

Proof. Apply the gauge transformation $z = e^{-2x}F$ to (4) and expand. The resulting normal-form operator is exactly L_3. □

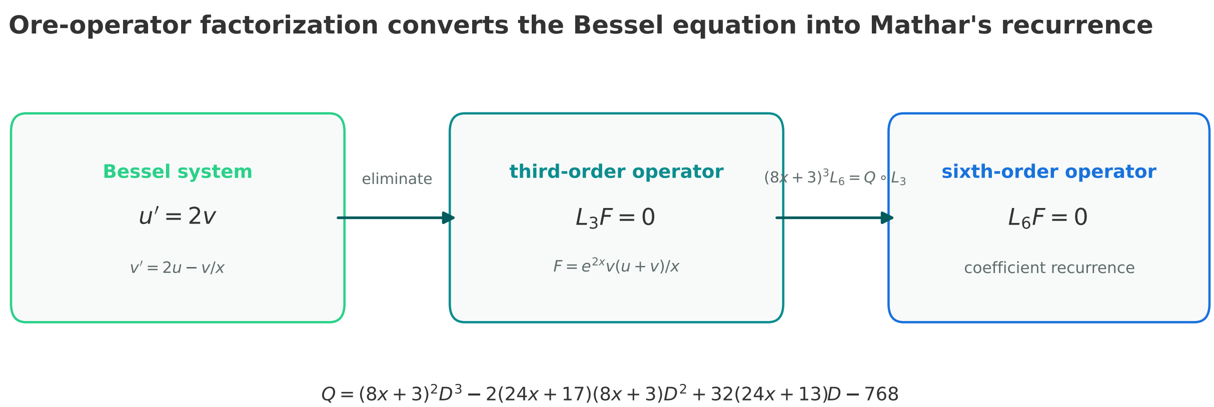

Figure 2: Proof architecture. The symmetric-square Bessel system produces a third-order annihilator. The exact Ore factorization then implies the sixth-order equation whose EGF coefficients are Mathar's recurrence.

4 Ore factorization and proof of the recurrence

Define the sixth-order differential operator

$$\begin{aligned} L_6 = {} & x^2\mathrm{D}^6 - 2x(6x-7)\mathrm{D}^5 + 2(16x^2 - 70x + 21)\mathrm{D}^4 \\ & + 4(12x^2 + 78x - 83)\mathrm{D}^3 - 8(18x^2 - 30x - 73)\mathrm{D}^2 \\ & - 192(3x-1)\mathrm{D} - 288, \end{aligned} \tag{6}$$

and the third-order left factor

$$Q = (8x+3)^2\mathrm{D}^3 - 2(24x+17)(8x+3)\mathrm{D}^2 + 32(24x+13)\mathrm{D} - 768. \tag{7}$$

Proposition 4.1 (Exact Ore factorization). *In the differential-operator ring* $\mathbb{Q}[x]\langle \mathrm{D}\rangle$, *with* $\mathrm{D}x = x\mathrm{D} + 1$,

$$(8x+3)^3 L_6 = Q \circ L_3. \tag{8}$$

Consequently $L_6 F = 0$.

Proof. To multiply operators in normal form, use

$$(a(x)\mathrm{D}^i) \circ (b(x)\mathrm{D}^j) = \sum_{k=0}^{i} \binom{i}{k} a(x) b^{(k)}(x) \mathrm{D}^{i-k+j}.$$

Applying this identity to (7) and the operator in proposition 3.2, then collecting the coefficients of $\mathrm{D}^0, \dots, \mathrm{D}^6$, gives exactly $(8x+3)^3$ times (6). This proves (8). Since $L_3 F = 0$, the right side annihilates F, and the factor $(8x+3)^3$ has nonzero constant term and is invertible in $\mathbb{Q}[[x]]$; hence $L_6 F = 0$ as a formal power series identity. ☐

Proof of theorem 1.1. For $F(x) = \sum_{n \ge 0} a_n x^n / n!$, the coefficient-transfer rule is

$$\left[\frac{x^m}{m!}\right] x^p \mathrm{D}^q F = (m)_p \, a_{m-p+q}, \tag{9}$$

where $(m)_p = m(m-1)\cdots(m-p+1)$ and the expression is zero for $p > m$. Extract the coefficient of $x^{n-4}/(n-4)!$ from $L_6 F = 0$, apply (9) term by term, and simplify. The result is exactly (1). Therefore Mathar's recurrence holds for every $n \ge 4$. ☐

5 A lower-order companion recurrence

The third-order annihilator itself gives another recurrence of the same width.

Corollary 5.1. *For every $n \ge 4$,*

$$\begin{aligned} 0 = {} & 3(n+2)(n+3)a_n + 2(4n^3 - 5n^2 - 37n - 16)a_{n-1} \\ & - 4(n-1)(12n^2 + 7n - 14)a_{n-2} \\ & - 8(n-1)(n-2)(4n-25)a_{n-3} \\ & + 192(n-1)(n-2)(n-3)a_{n-4}. \end{aligned} \tag{10}$$

Proof. Extract $x^{n-3}/(n-3)!$ from $L_3 F = 0$ using (9), then collect the five shifted terms. ☐

Remark 5.2. The coexistence of (1) and (10) is not contradictory. Both are polynomial-coefficient recurrences satisfied by the same D-finite sequence; (8) records an explicit operator relation between their differential counterparts.

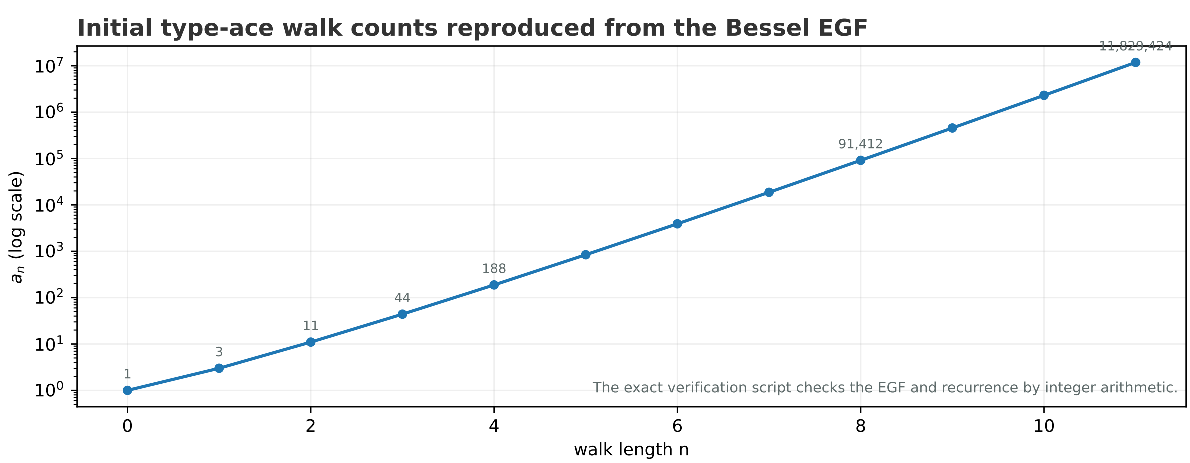

Figure 3: Initial values reconstructed independently from the coordinate-count formula. The plot is illustrative; all verification in the release uses exact integers.

6 Verification, scope, and review priorities

6.1 Exact independent checks

The accompanying script computes a_n directly from

$$a_n = \sum_{r+s+t=n} \binom{n}{r, s, t} A_r C_s E_t$$

using integer arithmetic. It reproduces the recorded initial values through $n = 80$ and verifies that the residuals of both (1) and (10) vanish over the full checked range. A separate symbolic routine implements the Ore multiplication rule in proposition 4.1 and proves coefficientwise that

$$(8x + 3)^3 L_6 - Q \circ L_3 = 0.$$

These calculations audit the displayed algebra; they are not substitutes for independent review of the proof or novelty claim.

Independent review target

A reviewer can check the manuscript in four short stages: (1) verify the three one-dimensional EGFs in lemma 2.1; (2) multiply the labelled EGFs to obtain (2); (3) eliminate the 3×3 rational system (5); and (4) expand the Ore product (8) and extract coefficients. The principal non-algebraic questions are literature priority and whether an equivalent recurrence proof exists under different terminology.

6.2 Literature status and limitations

Dershowitz introduced and enumerated the coordinate-type framework underlying the sequence. The OEIS entry inspected on 16 July 2026 still labels Mathar's recurrence conjectural. Searches for the exact sequence identifier, recurrence, and phrase "type ace" did not locate a later proof. Such searches can miss unpublished work, correspondence, non-indexed sources, or equivalent formulas written in another form.

The theorem proves the stated recurrence. It does not claim that L_3 or L_6 is a minimal annihilator, that (1) is the shortest recurrence possible, or that the Bessel factorization is the first possible derivation of the sequence.

Priority and review caveat

The formal identities are exact, but a targeted web and bibliographic search is not an exhaustive priority search. Equivalent proofs may exist in unpublished notes, computational files, correspondence, or literature indexed under different terminology. This work should be cited as a candidate resolution until independent experts confirm both the argument and its novelty.

7 Declarations and reproducibility

Institutional and computational-research disclaimer

This manuscript is an exploratory research preprint issued by the Metriq PRISM Laboratory, an interdisciplinary research laboratory of Metriq Foundation, Inc. PRISM—the Program for Research in Intelligent Systems and Methods—conducts research involving computational, mathematical, scientific, and machine-assisted methods. The work was developed as part of PRISM's experimental research program in computationally assisted mathematics. Metriq Foundation, acting through the PRISM Laboratory, directed the research process, evaluated the resulting argument, determined the form and scope of the manuscript, and accepts responsibility for its publication as a candidate result. Generative artificial intelligence and other computational tools were used as research instruments to support construction search, symbolic and numerical analysis, literature review, proof development, verification, and manuscript preparation. Their use does not constitute independent validation of the results presented. This manuscript has not yet undergone independent peer review and should not be regarded as an established resolution of the stated problem unless and until its arguments have been examined and confirmed by qualified mathematicians. Independent verification, criticism, replication, and identification of errors are expressly invited.

Data, code, funding, and competing interests

No empirical dataset was used. The source package contains the complete LuaLaTeX manuscript, formula-derived figures, the exact verification script, recorded output, build instructions, and release-quality-control records. This work received no external funding specifically designated for the result. Metriq Foundation, Inc. declares no competing financial interest in the mathematical conclusion.

Suggested citation

Metriq PRISM Laboratory, *A Bessel-Factorization Proof of Mathar's Recurrence for Type-ace Lattice Walks*, Metriq Foundation Research Preprint MF-MATH-2026-09, Version 1.0, 2026.

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